map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >

string

weak_ptr< smacc2::introspection
::SmaccStateMachineInfo >

weak_ptr< smacc2::introspection
::SmaccStateInfo >

vector< smacc2::introspection
::SmaccTransitionInfo >

vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >

type_info *

std::enable_shared
_from_this< SmaccStateInfo >

map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >

int

SharedPtr

SharedPtr

vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >

SMRunMode

int64_t

SharedPtr

vector< smacc2::ISmaccState * >

shared_ptr< smacc2
::introspection::SmaccState
Info >

SmaccStatus

map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >

shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >

recursive_mutex

atomic< StateMachineInternal
Action >

map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >

map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >

+stateReactorsInfo_

+eventGeneratorsInfo_

+demangledStateName
+fullStateName

+stateMachine_

+parentState_

+transitions_

+children_

+tid_

+staticBehaviorInfo_

+depth_
+stateIndex_

-reactorWaitDepth_

#int_

#eventsLogPub_
#stateMachinePub_

#stateMachineStatus
Pub_
#transitionHistory
Service_
#transitionLogPub_

-transitionLogHistory_

-runMode_

-stateSeqCounter_

#timer_

#currentState_

#currentStateInfo_

#status_msg_

#orthogonals_

#stateMachineInfo_

-eventQueueMutex_
-m_mutex_

-stateMachineCurrentAction

-stateCallbackConnections

-globalData_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

smacc2::introspection
::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

shared_ptr< rclcpp
::Logger >

vector< std::shared
_ptr< smacc2::ISmaccState
Reactor > >

vector< std::shared
_ptr< smacc2::ISmaccEventGenerator > >

atomic< bool >

#node_

vector< std::vector
< smacc2::ISmaccUpdatable * > >

-statusPub_

SmaccFifoScheduler *

processor_handle

thread

ExecutionModel

vector< smacc2::ISmaccUpdatable * >

double

vector< std::shared
_ptr< smacc2::ISmaccClient > >

vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >

mutex

#logger_

#stateReactors_

#eventGenerators_

#stateInfo_

-end_
-initialized_
-isInitialized_

-updatableStateElements_

-scheduler_

-processorHandle_

-signalDetectorThread_

-executionModel_

-updatableClients_

-signalDetector_

-smaccStateMachine_

-loggerPtr_

-stateMachine_

#clients_

-clientBehaviors_

-mutex_

smacc2::ISmaccState

+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >

smacc2::ISmaccOrthogonal

+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
+ onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()

map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >

-stateMachine_

-stateMachine_

-currentState

-stateMachine_

-currentOrthogonal_

-orthogonal_

#components_

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

-isShutdownRequested_

smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >

boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >

< void()>

#owner_

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

-markerScale_
-publishRate_

-markersMutex_

function< void()>

optional< std::future
< int > >

smacc2::SmaccSignal
< void()>

Options

shared_ptr< moveit
::planning_interface
::MoveGroupInterface >

shared_ptr< moveit
::planning_interface
::PlanningSceneInterface >

vector< cl_moveit2z
::TrajectoryHistoryEntry >

shared_ptr< tf2_ros
::TransformListener >

weak_ptr< tf2_ros::
Buffer >

weak_ptr< tf2_ros::
TransformListener >

mutex

shared_ptr< tf2_ros
::Buffer >

optional< rclcpp::Duration >

optional< rclcpp::Time >

-postFailureEventFn_
-postFinishEventFn_
-postSuccessEventFn_

-onEntryThread_
-onExitThread_

-onFailure_
-onFinished_
-onSuccess_

-postEventMotionExecution
Failed_
-postEventMotionExecution
Succeeded_

-onFailed_
-onSucceeded_

-options_

+moveGroupClientInterface

+planningSceneInterface

-trajectoryHistory_

-tfListener_

-sharedTfBuffer_

-sharedTfListener_

-tfMutex_

-tfBuffer_

-periodDuration_

-lastUpdate_

-m_mutex_

smacc2::SmaccAsyncClient
Behavior

+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

cl_moveit2z::CpMoveGroup
Interface

+ CpMoveGroupInterface()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ postEventMotionExecution
Succeeded()
+ postEventMotionExecution
Failed()
+ onMotionExecutionSucceeded()
+ onMotionExecutionFailed()

cl_moveit2z::CpTrajectory
History

+ getLastTrajectory()
+ getLastTrajectory()
+ pushTrajectory()

SharedPtr

cl_moveit2z::CpTfListener

+ CpTfListener()
+ ~CpTfListener()
+ onInitialize()
+ lookupTransform()
+ transformPose()
+ canTransform()
+ getBuffer()

float

Time

smacc2::ISmaccUpdatable

+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()

MarkerArray

bool

-cpMoveGroupInterface_

-cpMoveGroupInterface_

-cpMoveGroupInterface_

-trajectoryHistory_

-tfClient_

-markersPub_

-markerColor_

-lastPublishTime_

-markers_

-enabled_
-markersReady_

cl_moveit2z::CpMotionPlanner

+ CpMotionPlanner()
+ ~CpMotionPlanner()
+ onInitialize()
+ planToPose()
+ planToJointTarget()
+ planCartesianPath()
+ getCurrentState()
- applyPlanningOptions()

cl_moveit2z::CpTrajectory
Executor

+ CpTrajectoryExecutor()
+ ~CpTrajectoryExecutor()
+ onInitialize()
+ execute()
+ executePlan()
+ cancel()
+ getTrajectoryHistory()
- applyExecutionOptions()
- recordToHistory()

cl_moveit2z::CpJointSpace
TrajectoryPlanner

+ CpJointSpaceTrajectoryPlanner()
+ ~CpJointSpaceTrajectoryPlanner()
+ onInitialize()
+ planFromWaypoints()

cl_moveit2z::CpTrajectory
Visualizer

+ CpTrajectoryVisualizer()
+ ~CpTrajectoryVisualizer()
+ onInitialize()
+ update()
+ setTrajectory()
+ clearMarkers()
+ setColor()
+ setScale()
+ setEnabled()
+ isEnabled()

-postMotionExecutionFailure
Events

-cpMoveGroup_

-cpTrajectoryHistory_

-cpMotionPlanner_

-cpTrajectoryExecutor_

-tkJoy-
markersPub_

-cpJointSpaceTrajectoryPlanner_

-cpTrajectoryVisualizer_

-cpbehaviorMarkers_

-autocleanmarkers

cl_moveit2z::CbMoveit2zClient
BehaviorBase

+ onStateOrthogonalAllocation()
+ ~CbMoveit2zClientBehavior
Base()
# postMotionSuccess()
# postMotionFailure()

optional< std::string >

optional< bool >

vector< geometry_msgs
::msg::PoseStamped >

function< void(moveit
_msgs::msg::RobotTrajectory &)>

+group_
+ptp_link_

+allowInitialTrajectoryState
JointDiscontinuity_

-endEffectorTrajectory_

-postIncorrectInitialState
Event_
-postJointDiscontinuity
Event_

cl_moveit2z::CbMoveEndEffector
Trajectory

+ CbMoveEndEffectorTrajectory()
+ CbMoveEndEffectorTrajectory()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onExit()
# computeJointSpaceTrajectory()
# executeJointSpaceTrajectory()
# generateTrajectory()
# createMarkers()
# getCurrentEndEffectorPose()
- initializeROS()